From flux to divergence

Start with a fixed closed surface and free charge.

$$\oint_S D \cdot da = \int_V \rho_f \, dV$$

Divergence theorem converts the surface flux to a volume integral.

$$\int_V (\nabla \cdot D - \rho_f)\, dV = 0$$

An arbitrary volume in a smooth bulk region gives:

$$\nabla \cdot D = \rho_f$$

$$\oint_S B \cdot da = 0 \quad \Longrightarrow \quad \nabla \cdot B = 0$$

At an interface, return to the integral laws.

From circulation to curl

Use a fixed contour; its surface normal follows the right-hand rule.

$$\oint_C E \cdot dl = -\frac{d}{dt}\int_A B \cdot da$$

$$\int_A \left(\nabla \times E + \frac{\partial B}{\partial t}\right) \cdot da = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Apply Stokes in the same way to Ampere-Maxwell law:

$$\oint_C H \cdot dl = \int_A \left(J_f + \frac{\partial D}{\partial t}\right) \cdot da$$

$$\nabla \times H = J_f + \frac{\partial D}{\partial t}$$

These local equations hold in smooth bulk regions.

Charge conservation is compulsory

$$\nabla \cdot (\nabla \times H) = 0$$

$$0 = \nabla \cdot J_f + \frac{\partial}{\partial t}(\nabla \cdot D)$$

$$\nabla \cdot D = \rho_f$$

$$\frac{\partial \rho_f}{\partial t} + \nabla \cdot J_f = 0$$

Integrating over a fixed volume gives the charge balance:

$$\frac{d}{dt}\int_V \rho_f \, dV = -\oint_S J_f \cdot da$$

Outward current reduces the free charge in the volume.

Omitting displacement current would force div J = 0 everywhere.

The capacitor closes the current loop

Parallel plates; vacuum gap; uniform field; ignore fringing.

$$D_n = \frac{Q}{A}, \qquad E_n = \frac{V}{d}$$

$$I_d = \int_A \frac{\partial D}{\partial t} \cdot da = A \frac{\partial D_n}{\partial t}$$

$$I_d = \frac{dQ}{dt} = \frac{\epsilon_0 A}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

A = 0.010 m^2; d = 1.0 mm; dV/dt = 10^6 V/s.

$$C = 8.85 \times 10^{-11}\ \mathrm{F}$$

$$I_d = 8.85 \times 10^{-5}\ \mathrm{A} = 88.5\ \mu\mathrm{A}$$

The gap carries displacement current, not free-charge transport.

Separate field definitions from material models

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

Assume linear, local, isotropic, nondispersive response.

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \qquad \mathbf{M} = \chi_m \mathbf{H}$$

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}$$

$$\mathbf{B} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu \mathbf{H}$$

$$\mathbf{J}_{cond} = \sigma \mathbf{E}$$

Free current may also contain impressed sources.

For anisotropy use a tensor; for dispersion include time memory.

Normal D: the charged pillbox

Normal n points from medium 1 to medium 2. Face area is A.

$$\oint_S D \cdot da = Q_{f,enc}$$

$$(D_{2n} - D_{1n})A + \Phi_{side} = \sigma_f A + Q_{f,bulk}$$

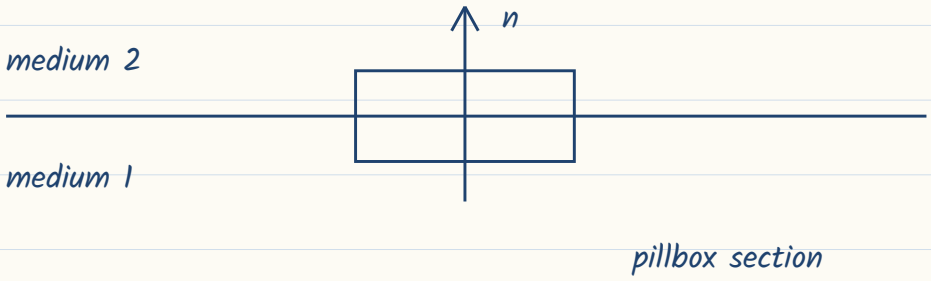
As height h tends to zero, finite side flux and bulk charge vanish.

$$D_{2n} - D_{1n} = \sigma_f$$

$$\hat{n} \cdot (D_2 - D_1) = \sigma_f$$

$$\epsilon_2 E_{2n} - \epsilon_1 E_{1n} = \sigma_f$$

The last line assumes scalar permittivities. Free charge controls D.



pillbox section

Normal B: the unbroken flux

$$\oint_S B \cdot da = 0$$

$$(B_{2n} - B_{1n})A + \Phi_{side} = 0$$

Shrink the same pillbox. The side contribution tends to zero.

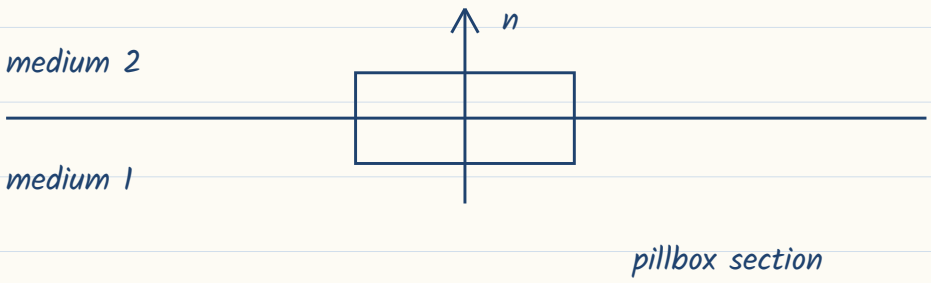
$$B_{2n} = B_{1n}$$

$$\hat{n} \cdot (B_2 - B_1) = 0$$

$$\mu_2 H_{2n} = \mu_1 H_{1n}$$

Normal H need not be continuous when permeability changes.

This is the absence of magnetic charge in ordinary Maxwell theory.



pillbox section

Tangential E: shrink the loop

Long sides have length L; short sides have height h.

$$\oint_C E \cdot dl = -\frac{d}{dt}\int_A B \cdot da$$

$$(E_{2t} - E_{1t})L + O(h) = O(hL)$$

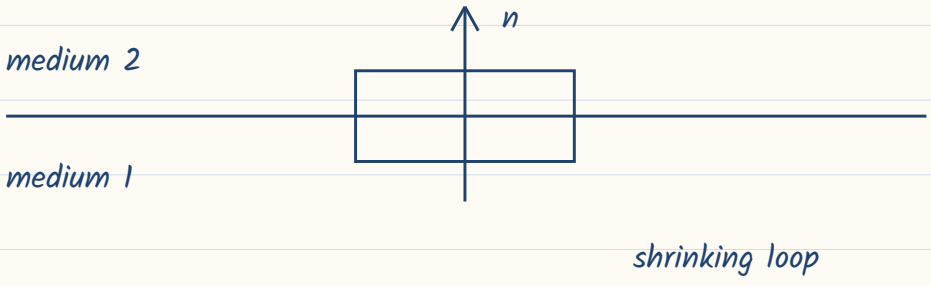
Finite time derivative of B makes the flux term vanish as h -> 0.

$$E_{2t} = E_{1t}$$

$$\hat{n} \times (E_2 - E_1) = 0$$

This remains true during induction at an ordinary interface.

Exclude singular magnetic sheets and ideal dipole-layer models.



Tangential H: retain the sheet current

$$\oint_C H \cdot dl = \int_A J_f \cdot da + \frac{d}{dt}\int_A D \cdot da$$

Bulk current and regular displacement flux vanish with loop height.

A free sheet current remains finite; loop orientation fixes its sign.

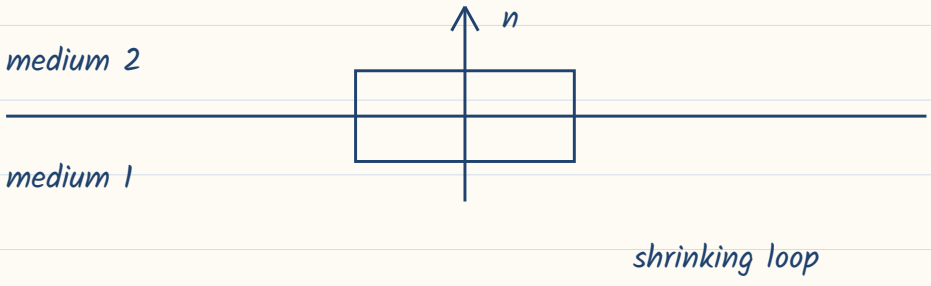
$$\hat{n} \times (H_2 - H_1) = K_f$$

$$H_{2t} - H_{1t} = K_f \times \hat{n}$$

$$\hat{n} = \hat{z}, \quad K_f = K\hat{x} \quad \Rightarrow \quad \Delta H_t = -K\hat{y}$$

$$\hat{z} \times (-K\hat{y}) = K\hat{x}$$

Always perform this Cartesian sign check before substitution.



Back to the electrostatic potential

$$E = -\nabla\Phi, \quad D = \epsilon E, \quad \nabla \cdot D = \rho_f$$

$$\nabla \cdot (\epsilon \nabla \Phi) = -\rho_f$$

Only uniform scalar epsilon permits the familiar Poisson form:

$$\nabla^2 \Phi = -\rho_f / \epsilon$$

At an ordinary interface with no ideal dipole sheet:

$$\Phi_1 = \Phi_2$$

$$\epsilon_1 \frac{\partial \Phi_1}{\partial n} - \epsilon_2 \frac{\partial \Phi_2}{\partial n} = \sigma_f$$

Both normal derivatives use the same direction from 1 to 2.

Two interface calculations

Uncharged dielectric: eps1 = 2 eps0; eps2 = 5 eps0.

$$\mathbf{E}_1 = 30\hat{x} + 50\hat{z}\ \text{V/m}, \quad \hat{n} = \hat{z}$$

$$E_{2x} = 30, \quad E_{2z} = \frac{\epsilon_1}{\epsilon_2} 50 = 20\ \text{V/m}$$

$$\mathbf{E}_2 = 30\hat{x} + 20\hat{z}\ \text{V/m}$$

Current sheet: Kf = 6 x-hat A/m; mu2 = 2 mu1.

$$\mathbf{H}_1 = 2\hat{y} + 3\hat{z}\ \text{A/m}$$

$$\Delta H_t = -6\hat{y}, \quad H_{2z} = 3/2$$

$$\mathbf{H}_2 = -4\hat{y} + 1.5\hat{z}\ \text{A/m}$$

The wave equation follows

Homogeneous, source-free, lossless scalar medium; epsilon, mu constant.

$$\nabla \times E = -\mu \frac{\partial H}{\partial t}$$

$$\nabla \times (\nabla \times E) = -\mu\epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\nabla \times (\nabla \times E) = \nabla(\nabla \cdot E) - \nabla^2 E$$

$$\nabla \cdot D = 0 \quad \Rightarrow \quad \nabla \cdot E = 0$$

$$\nabla^2 E - \mu\epsilon \frac{\partial^2 E}{\partial t^2} = 0$$

$$v = 1/\sqrt{\mu\epsilon}$$

The next lecture uses interface conditions to obtain reflection.